

eSim Semester Long Internship - Autumn 2026



**Project Title: Autonomous Ambient Light-Controlled
Lighting System**

Software Used: eSim v2.3

**Layer Count: 4-Layer Printed Circuit Board (F.Cu,
In1.Cu, In2.Cu, B.Cu)**

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Contents

Project Overview	3
Circuit Description	3
Schematic	5
Footprint Assignment	5
PCB Layout	6
Routing Methodology.....	7
DRC Results	8
3D View.....	9
Design Considerations & Conclusion.....	12

Project Overview

An **Autonomous Ambient Light-Controlled Lighting System** is an automated lighting management circuit designed to switch an output load based on ambient light conditions. Using an LDR light-sensor network alongside dual LM741 operational amplifiers, the system compares ambient light levels against pre-set threshold voltages to trigger switching transistors and status indicators.

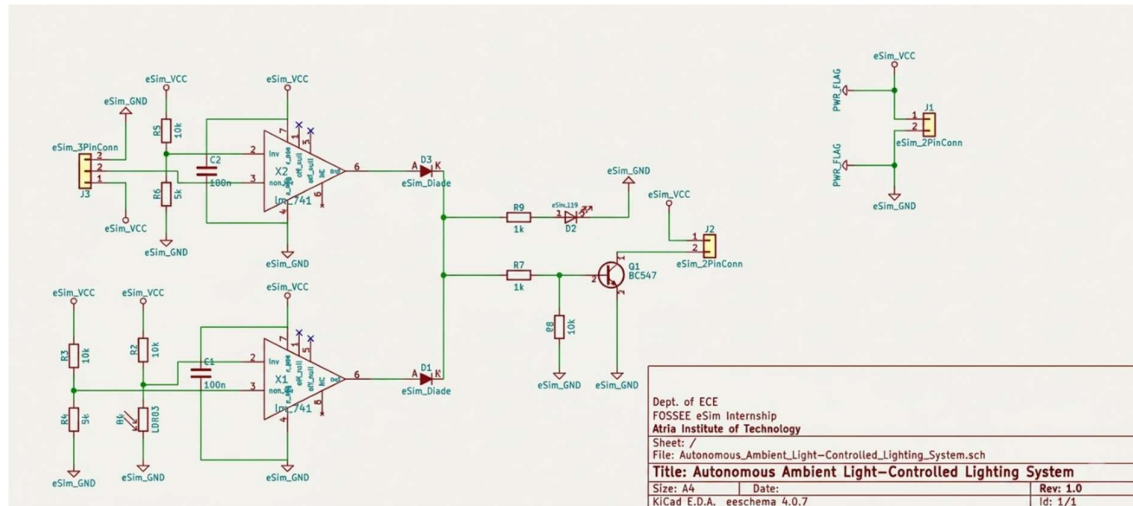
Circuit Description

The circuit consists of five primary functional stages based on the schematic:

- **Power Entry Stage (J1 & Power Flags):** Main power supply input is connected via 2-pin connector J1 (eSim_2PinConn). Global power supply rails are defined by net flags eSim_VCC (Pin 1) and eSim_GND (Pin 2).
- **Sensor & Reference Voltage Stage (X1 Op-Amp Inputs):**
 - **LDR Sensor Input:** Resistor R2 (10 k Ω) and Light Dependent Resistor R1 (LDR03) form a voltage divider connected to non-inverting terminal Pin 3 of operational amplifier X1 (lm_741).
 - **Reference Threshold:** Resistors R3 (10 k Ω) and R4 (5 k Ω) form a fixed reference voltage divider connected to inverting terminal Pin 2 of X1.

- Filtering: Ceramic capacitor C1 (100 nF) is connected across pins 2 and 3 of X1 to suppress high-frequency noise.
- Auxiliary Signal Stage (X2 Op-Amp Inputs):
 - Fixed Reference: Resistors R5 (10 k Ω) and R6 (5 k Ω) form a fixed reference voltage divider connected to inverting terminal Pin 2 of X2.
 - External Input Terminal J3: 3-pin connector J3 (eSim_3PinConn) connects directly to non-inverting terminal Pin 3 of X2.
 - Filtering: Capacitor C2 (100 nF) filters high-frequency noise across pins 2 and 3 of X2.
- Diode-OR Logic Stage (D1, D3): Outputs (Pin 6) of both op-amps drive steering diodes D1 and D3 (eSim_Diode), implementing OR logic to allow either comparator output to trigger the switching stage.
- Output Switching Stage (Q1, D2, J2):
 - Transistor Driver: The combined diode output connects through resistor R7 (1 k Ω) and pull-down resistor R8 (10 k Ω) to the base of NPN transistor Q1 (BC547), which controls the load connected across 2-pin terminal J2.
 - Visual Indicator: Current-limiting resistor R9 (1 k Ω) drives status LED D2 (eSim_LED) for output visual confirmation

Schematic

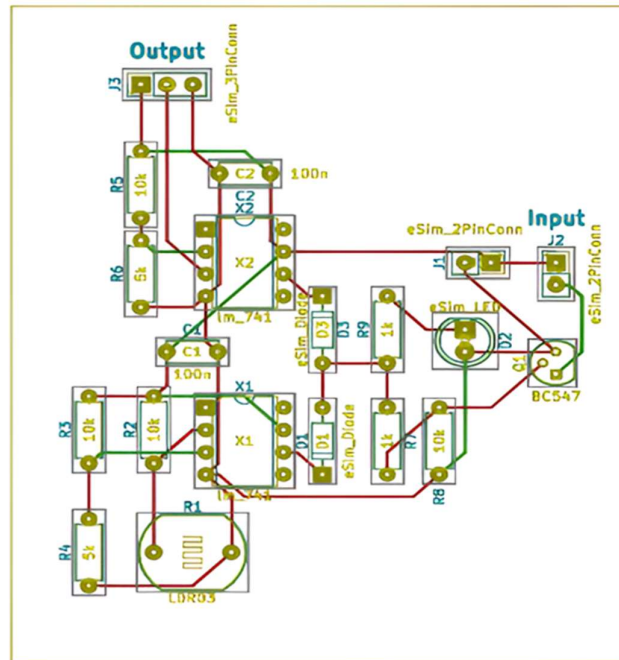


Footprint Assignment

1	C1 -	100n : Capacitors_THT:C_Disc_D5.0mm_W2.5mm_P5.00mm
2	C2 -	100n : Capacitors_THT:C_Disc_D5.0mm_W2.5mm_P5.00mm
3	D1 -	eSim_Diode : Diodes_THT:D_DO-35_SOD27_P7.62mm_Horizontal
4	D2 -	eSim_LED : LEDs_LED_D5.0mm
5	D3 -	eSim_Diode : Diodes_THT:D_DO-35_SOD27_P7.62mm_Horizontal
6	J1 -	eSim_2PinConn : Pin_Headers:Pin_Header_Straight_1x02_Pitch2.54mm
7	J2 -	eSim_2PinConn : Pin_Headers:Pin_Header_Straight_1x02_Pitch2.54mm
8	J3 -	eSim_3PinConn : Pin_Headers:Pin_Header_Straight_1x03_Pitch2.54mm
9	Q1 -	BC547 : TO_SOT_Packages_THT:TO-92_Molded_Narrow
10	R1 -	LDR03 : Opto-Devices:Resistor_LDR_10x8.5_RM7.6
11	R2 -	10k : Resistors_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal
12	R3 -	10k : Resistors_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal
13	R4 -	5k : Resistors_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal
14	R5 -	10k : Resistors_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal
15	R6 -	5k : Resistors_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal
16	R7 -	1k : Resistors_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal
17	R8 -	10k : Resistors_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal
18	R9 -	1k : Resistors_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal
19	X1 -	lm_741 : Housings_DIP:DIP-8_W7.62mm
20	X2 -	lm_741 : Housings_DIP:DIP-8_W7.62mm

Component Reference	Part Name	Schematic Symbol	Assigned PCB Footprint	Purpose
X1, X2	lm_741	Op-Amp	DIP-8_W7.62mm	Voltage Comparators
R1	LDR03	LDR	Resistor_LDR_10x8.5_RM7.6	Light Dependent Resistor Sensor
R2, R3, R5	10 k Ω	R	R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal	Pull-up & Divider Resistors
R4, R6	5 k Ω	R	R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal	Reference Voltage Divider Resistors
R7, R9	1 k Ω	R	R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal	Base Drive & LED Current Limiting
R8	10 k Ω	R	R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal	Base Pull-Down Resistor
C1, C2	100 nF	C	C_Disc_D5.0mm_W2.5mm_P5.00mm	Differential Noise Filter Capacitors
D1, D3	eSim_Diode	Diode	D_DO-35_SOD27_P7.62mm_Horizontal	Diode OR-Logic Steering
D2	eSim_LED	LED	LED_D5.0mm	Visual Output Status Indicator
Q1	BC547	NPN Transistor	TO-92_Molded_Narrow	Output Switching Transistor
J1, J2	eSim_2PinConn	2-Pin Connector	Pin_Header_Straight_1x02_Pitch2.54mm	Power Input & Output Header
J3	eSim_3PinConn	3-Pin Connector	Pin_Header_Straight_1x03_Pitch2.54mm	External Signal Header

PCB Layout



Dept. of ECE
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Sheet:
File: Autonomous_Ambient_Light-Controlled_Lighting_System.kicad_pcb

Title: Autonomous Ambient Light-Controlled Lighting System

Size: A4	Date:	Rev: 1.0
KiCad E.D.A. pcbnew 4.0.7		Id: 1/1

Routing Methodology

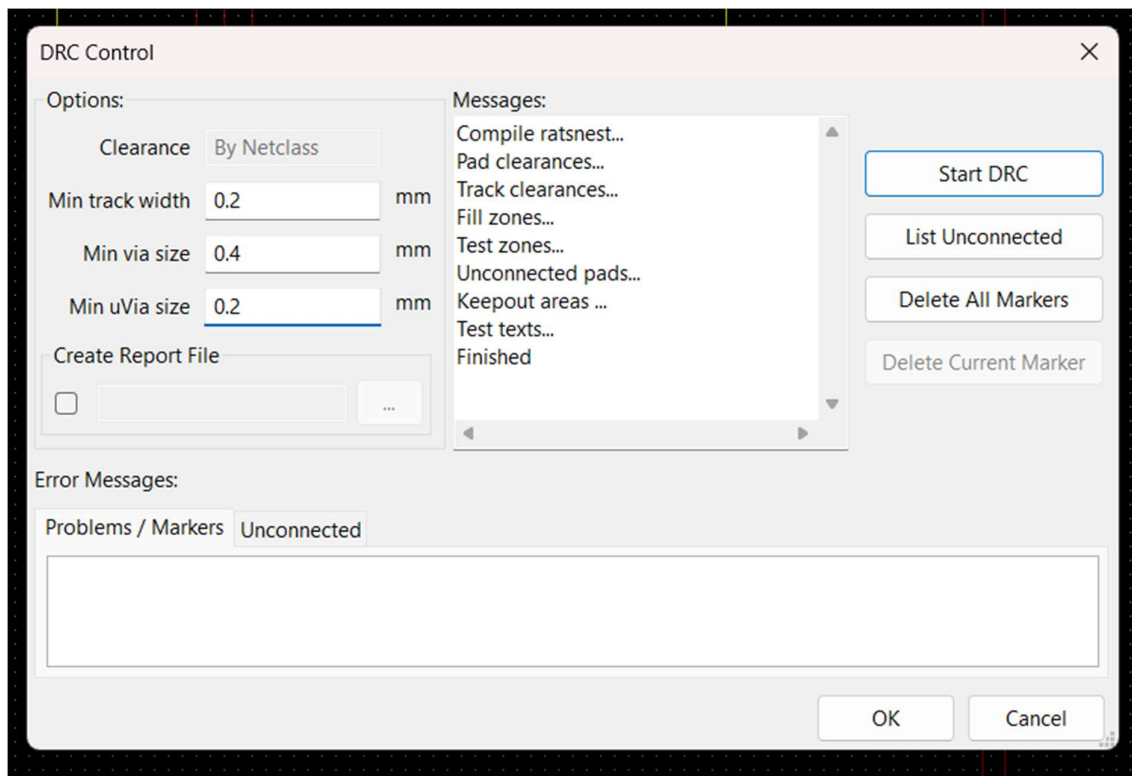
To ensure high signal integrity, minimal interference, and proper voltage distribution, a 4-layer PCB stackup strategy was implemented:

- Layer Stackup Configuration:
 - F.Cu (Top Layer): Dedicated to primary component interconnects and signal trace routing.
 - In1.Cu (Inner Layer 1): Solid ground plane (eSim_GND) pour to minimize loop inductance and provide EMI shielding.
 - In2.Cu (Inner Layer 2): Solid power plane (eSim_VCC) pour to maintain low-impedance power delivery across all IC pins.
 - B.Cu (Bottom Layer): Used for auxiliary signal routing and ground return copper fills.
- Silkscreen Identification: Clear silkscreen labels (Input near connector J2 and Output near connector J3) were placed on the front silkscreen layer (F.SilkS) for ease of physical identification.

DRC Results

A complete Design Rule Check (DRC) was performed in Pcbnew to ensure zero manufacturing violations:

- Unconnected Nets: 0 (links 36, nc 0, net 7: not conn 0)
- Clearance Violations: 0
- Short Circuits: 0



3D View

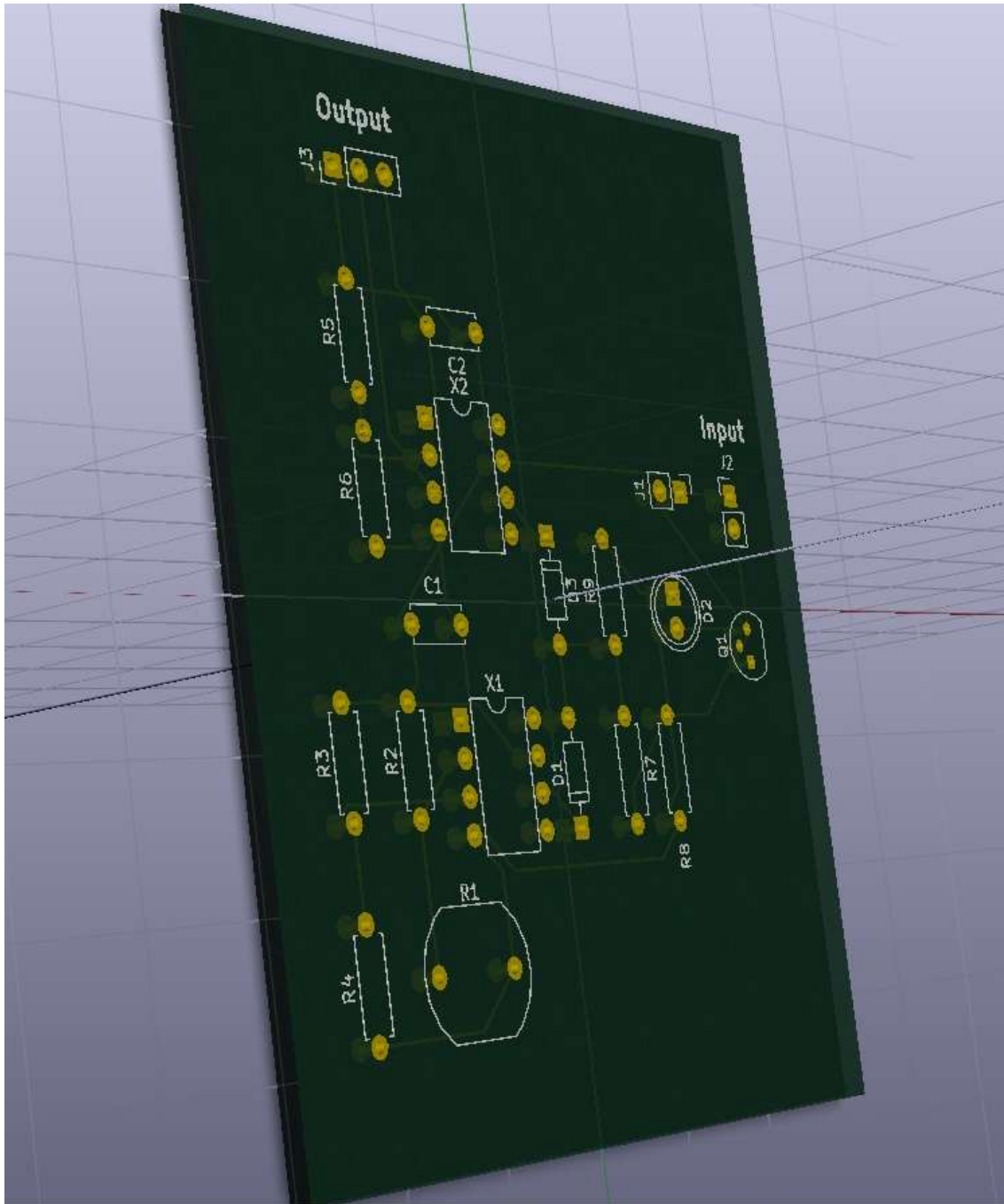


Fig: 3D isometric perspective view showing PCB substrate, board thickness, through-hole drill pads, and surface routing.

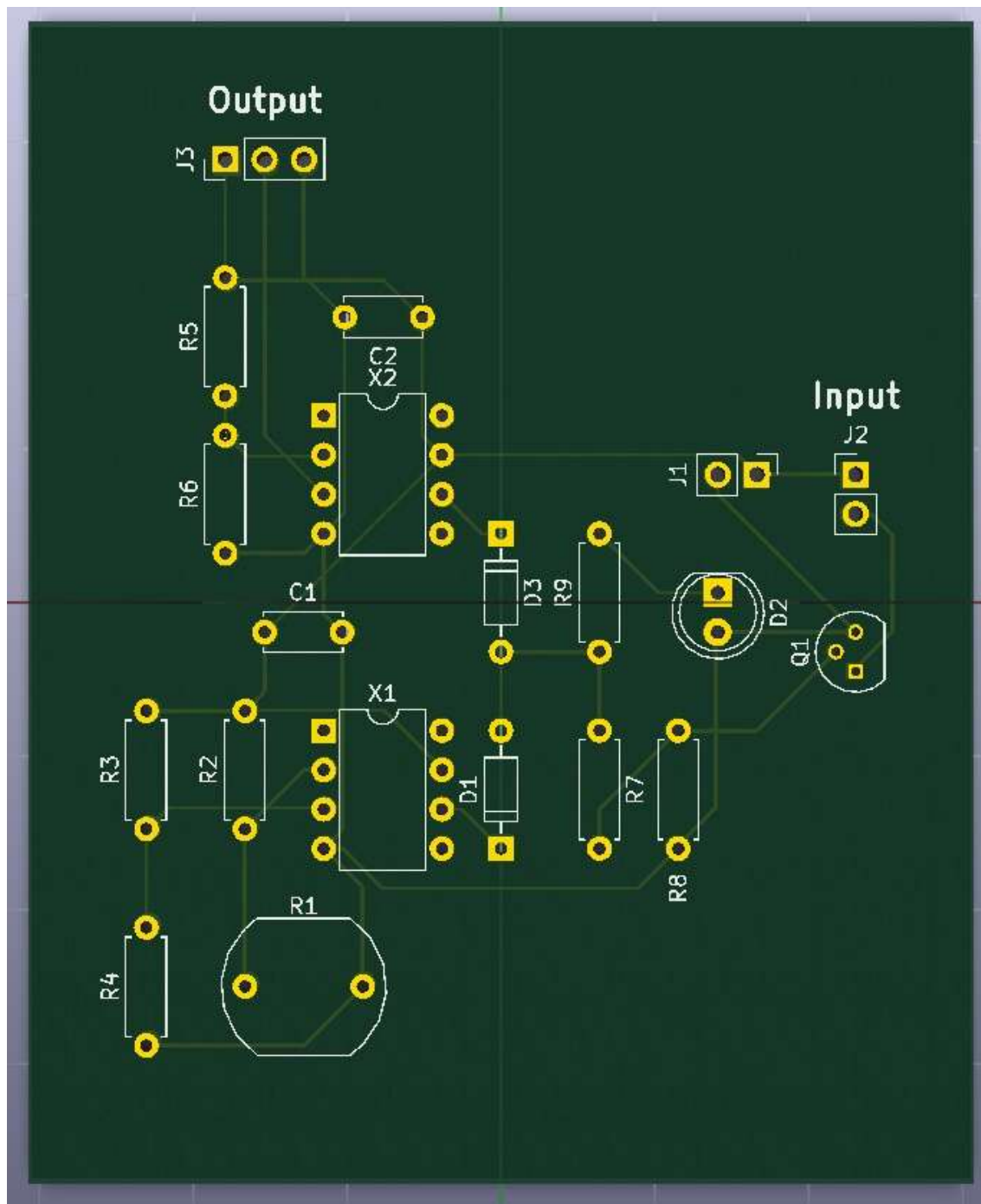


Fig: 3D top view showing silkscreen footprints, text labels, and component pad locations.

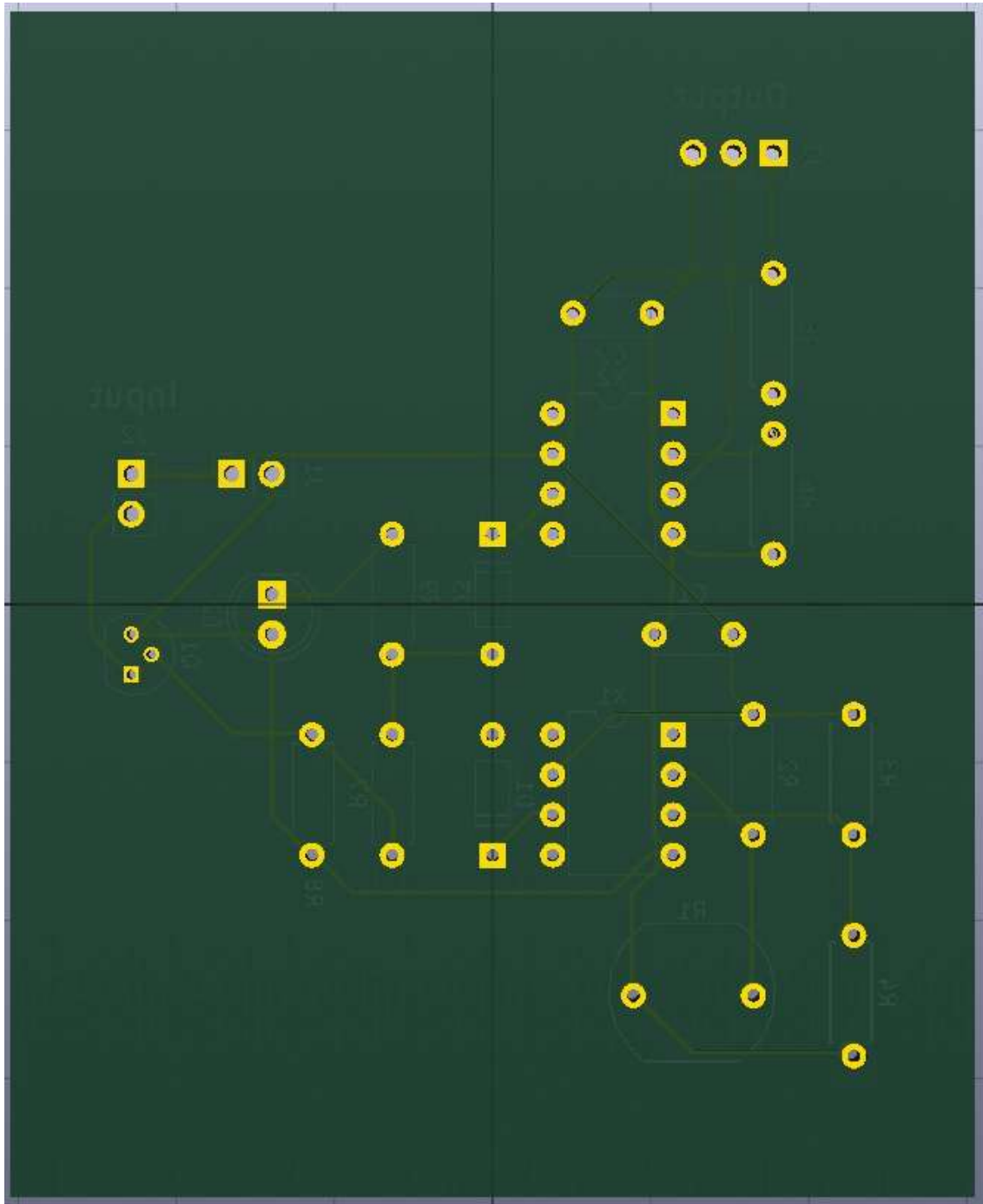


Fig: 3D bottom view showing bottom copper trace routing and terminal pads.

Design Considerations & Conclusion

- Superior Noise Mitigation: A dedicated 4-layer stackup with continuous internal Ground (In1.Cu) and Power (In2.Cu) planes drastically lowers loop inductance and eliminates supply ripple. Decoupling capacitors (C_1 , C_2) placed directly at the op-amp supply pins (X_1 , X_2) ensure pristine signal integrity and robust EMI suppression.
- Prototyping & Assembly Readiness: Footprints and silkscreen legends conform strictly to standard IPC tolerances. Highly visible reference designators (R1–R9, D1–D3, Q1, J1–J3) and explicit functional callouts (Input, Output) enable error-free manual soldering and automated assembly.
- Manufacturing Compliance: Design Rule Checking (DRC) verifies zero errors and zero unconnected items. Trace widths, pad clearances, and board outline dimensions meet all standard fabrication requirements, making the layout fully validated for Gerber file export.

The **Autonomous Ambient Light-Controlled Lighting System** PCB design is fully optimized and production-ready. Integrating multi-layer power distribution, signal isolation, and thermal/trace routing into a verified layout ensures 100% compliance with FOSSEE standards and physical fabrication readiness.